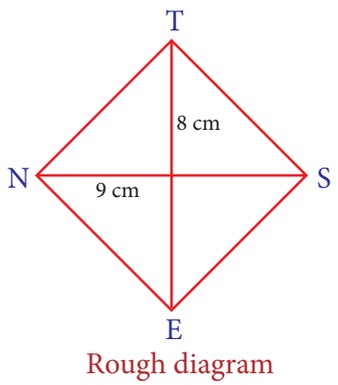
5.14.3 Construction of a rhombus when two diagonals are given

Example 5.36

Construct a rhombus NEST with NS = 9 cm and ET = 8 cm. Also find its area.



Rough diagram

Solution:

Given: NS = 9 cm and ET = 8 cm

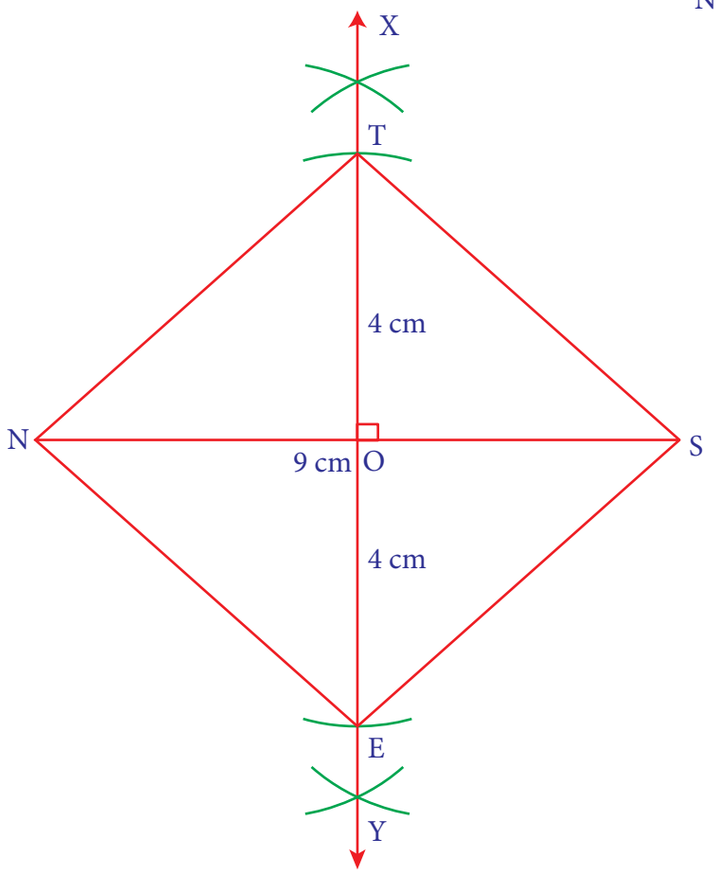


Fig. 5.55

Steps:

1. Draw a line segment NS = 9 cm.
2. Draw the perpendicular bisector XY to NS. Let it cut NS at O.
3. With O as centre, draw arcs of radius 4 cm on either side of O which cut OX at T and OY at E.
4. Join NE, ES, ST and TN.
5. NEST is the required rhombus.

Calculation of Area:

Area of rhombus NEST $= \frac{1}{2} \times d_1 \times d_2 \; sq.units$

$= \frac{1}{2} \times 9 \times 8 = 36 \; sq.cm$

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